ROADWATCH · OXHACK

Presentation Speech

Presenter 1 · about 4 min 30 s · slide-by-slide script

Event: OxHack · **Length:** about 4 minutes 30 seconds, leaving 30 seconds before the 5-minute limit · **Spoken words:** 555 (536 without the optional line)

Roles. Presenter 1 speaks throughout. Presenter 2 silently advances the slides (`RoadWatch_OxHack_Pitch.pptx`) and drives the live dashboard. The dashboard is open at the retrospective example week 16 March 2026. Presenter 2's matching actions are in `docs/final/demo_instructions.md`; both roles side by side are in `docs/final/combined_run_of_show.md`.

Delivery.

- Speak at a relaxed pace, about 130 words per minute.
- Pause briefly at each transition cue; the timestamps assume those pauses.
- If you are more than 15 seconds behind when the demo starts (later than 2:55), skip the line marked *(optional)*.
- If a full rehearsal runs past 4:40, also drop "and an offline rebuild reproduces every table byte for byte" (Slide 3) and "for block shapes and road attributes" (Slide 2). Both facts stay visible on the slides.
- Lines starting with `>` are the exact spoken words. Text in *italics* is a cue and is not spoken.

0:00 – 0:25 · Slide 1 · Problem

> Cities have far more streets than inspectors. Brooklyn has 23,448 street blocks, and in a typical week fewer than one in a hundred gets a pothole report. RoadWatch asks: can historical evidence show which blocks are most likely to get one next week, so a fixed inspection budget goes where it is needed?

Cue: nod to Presenter 2 → Slide 2.

0:25 – 0:55 · Slide 2 · Data collection

> We used only real public data: over seventy-one thousand Brooklyn pothole reports from NYC 311 since 2020, the city's official street centreline for block shapes and road attributes, and NOAA weather. The hard part is location: thirty-nine percent of reports name an intersection or a vague stretch of street, so they could belong to several blocks, typically four.

Cue → Slide 3.

0:55 – 1:27 · Slide 3 · Processing

> Every download is archived unchanged with checksums and validated, and an offline rebuild reproduces every table byte for byte. Reports are matched to blocks by street names and coordinates; ambiguous ones keep all their candidates. Weeks with too many unlocatable reports are marked unknown, never zero. And every feature is point-in-time: only 311 data published two days before the forecast, and weather from five days before.

Cue → Slide 4.

1:27 – 1:57 · Slide 4 · Prediction

> The model is LightGBM: gradient-boosted trees over 38 features, including recent report history, neighbouring blocks and road attributes like length and lanes. To test it honestly, we replayed history: sixteen monthly retrains from July 2025, each trained only on the past. The main comparison is a historical-rate baseline, which ranks blocks by how often they have been reported before.

Cue → Slide 5.

1:57 – 2:27 · Slide 5 · Evidence

> Across 67 held-out weeks with a 235-block list, LightGBM caught 13.4 percent of next week's reports, against 8.4 for the historical rate and 1 percent for a random list. Precision rose from 6.6 to 9.5 percent: a gain of 2.3 to 3.4 points, with 95 percent confidence. That is the strict target, where each report counts on exactly one block.

Cue → Slide 6.

2:27 – 2:40 · Slide 6 · Decision-making

> Turning ranks into decisions is deliberately simple: choose a weekly capacity, here 235 blocks, and take the top-ranked ones. Let's see it in the dashboard.

Cue: Presenter 2 switches to the dashboard (Alt+Tab). Keep talking.

2:40 – 3:43 · Live demo (dashboard)

> This is the week of 16 March 2026, a retrospective example, not a live recommendation. The blue lines are the 235 blocks to inspect, ranked in the table. If the budget doubles to 2 percent, the list grows to 469 blocks. Every block shows its reasons: 17th Avenue had 59 reports naming it or its end intersections in the past year, 18 in the last month. That is evidence, not proof of a defect. One subtlety: what counts as catching a report depends on what an inspection covers. If inspecting a block also covers its end intersections, spreading inspections across intersections looks best. Without that assumption the strict list wins, so it is our default. *(optional)* That alternative was also proposed after we had seen test results, so we do not treat it as confirmed.

Cue: Presenter 2 switches back to the slides and advances → Slide 7.

3:43 – 4:08 · Slide 7 · Limitations

> Now the limits. In March 2026 reports roughly quadrupled. The ranking held up, but our weekly volume forecast fell 42 percent short, so capacity stays a human decision. We predict reports, not confirmed defects, and reports reflect who complains as well as road condition. We claim no savings and no failures prevented.

Cue → Slide 8.

4:08 – 4:30 · Slide 8 · Real-world improvement and close

> To make this deployable, we need inspection outcomes to learn real defects, better intersection locations, and a prospective trial that freezes the model and scores future weeks. RoadWatch shows that public data can rank where to look first: honestly measured, and ready for a field test. Thank you.

End cue: Presenter 1 says "Thank you" and stops. Presenter 2 leaves Slide 8 on screen for questions.

If the live demo fails (Presenter 2 raises a hand and stays on Slide 6)

Replace the demo lines with these (about 45 seconds), then continue with Slide 7:

> "Here is the same week in our dashboard, on the left: the blue lines are the 235 blocks to inspect, ranked in the table beside the map. Each listed block comes with its report history and the model's main drivers, and the list can be exported for inspectors. On the right is one honest subtlety: what counts as catching a report depends on what an inspection covers. If inspecting a block also covers its end intersections, the non-adjacent lists reach 17.3 percent; without that assumption the strict list is best, at 13.5 percent, so it is our default."

Timing check

Segment	Planned window	Words	At 130 wpm	Synthetic voice (104 wpm)
Slide 1 · Problem	0:00–0:25	53	24 s	28 s
Slide 2 · Data	0:25–0:55	58	27 s	33 s
Slide 3 · Processing	0:55–1:27	66	30 s	37 s
Slide 4 · Prediction	1:27–1:57	59	27 s	35 s
Slide 5 · Evidence	1:57–2:27	60	28 s	37 s
Slide 6 · Decision	2:27–2:40	25	12 s	15 s
Live demo	2:40–3:43	134 (115 without optional)	62 s	75 s
Slide 7 · Limitations	3:43–4:08	52	24 s	33 s
Slide 8 · Close	4:08–4:30	48	22 s	25 s

Segment	Planned window	Words	At 130 wpm	Synthetic voice (104 wpm)
Total	**4:30**	**555**	**4:16 + pauses**	**5:19**

How this was timed. Every segment was read aloud by the Windows speech synthesiser and measured (`scripts/final/speech_timing.py` writes `outputs/final/verification/speech_timing.json`). That voice reads slowly, at 104 words per minute, so its 5:19 total is a worst case. At a normal presenting pace of 130 words per minute the words take 4:16; with about 1.5 seconds of pause at each of the 9 transitions, the plan lands at 4:30. That leaves 30 seconds of contingency before 5:00.

A synthetic voice is not a human rehearsal. Rehearse once with a stopwatch:

- Check that you reach Slide 6 by about 2:35.
- If you are slower, apply the trims listed under *Delivery*.

Numbers used, all from saved evaluation files

Number	Meaning	Source
23,448	eligible Brooklyn street blocks	`outputs/stage3/matching_diagnostics.json`
"fewer than one in a hundred"	0.9% of block-weeks positive (strict target, test period)	`outputs/stage4/test_results.csv`
"over seventy-one thousand"	71,609 Brooklyn pothole reports	`matching_diagnostics.json`
39 percent, typically four	39.3% ambiguous; median 4 candidate blocks	`matching_diagnostics.json`
38 features, sixteen retrains	selected feature set; retrain schedule	`data/processed/stage4/selection.json`, model metadata
67 weeks, 235 blocks, 13.4% vs 8.4% vs 1%	strict-target top-1% report capture: LightGBM vs historical rate vs random	`outputs/stage4/test_results.csv`
6.6% → 9.5%; +2.3 to +3.4 points	precision@1% and its paired bootstrap 95% CI	`outputs/stage4/bootstrap_comparisons.json`
469 blocks at 2%	ceil(0.02 × 23,448)	dashboard
17th Avenue: 59 reports, 18 in the last month	rank 1 on 16 Mar 2026 (strict list)	dashboard worklist
quadrupled; 42% short	March mean 843 vs 209 per week; forecast bias −42%	`outputs/stage5/volume_forecast_results.json`